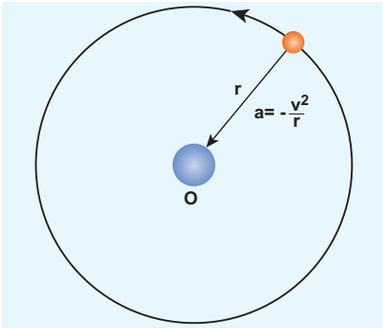


Figure 6.6 Point mass orbiting in a circular orbit.

Here v is the velocity and r, the distance of the planet from the center of the orbit (Figure 6.6).

The velocity in terms of known quantities r and T, is

$$v = \frac{2\pi r}{T} \qquad (6.5)$$

Here T is the time period of revolution of the planet. Substituting this value of v in equation (6.4) we get,

$$a = -\frac{\left(\frac{2\pi r}{T}\right)^2}{r} = -\frac{4\pi^2 r}{T^2} \qquad (6.6)$$

Substituting the value of 'a' from (6.6) in Newton's second law, $F = ma$, where 'm' is the mass of the planet.

$$F = -\frac{4\pi^2 mr}{T^2} \qquad (6.7)$$

From Kepler's third law,

$$\frac{r^3}{T^2} = k\left(constant\right) \qquad (6.8)$$

$$\frac{r}{T^2} = \frac{k}{r^2} \qquad (6.9)$$

By substituting equation 6.9 in the force expression, we can arrive at the law of gravitation.

$$F = -\frac{4\pi^2 mk}{r^2} \qquad (6.10)$$

Here negative sign implies that the force is attractive and it acts towards the center. In equation (6.10), mass of the planet 'm' comes explicitly. But Newton strongly felt that according to his third law, if Earth is attracted by the Sun, then the Sun must also be attracted by the Earth with the same magnitude of force. So he felt that the Sun's mass (M) should also occur explicitly in the expression for force (6.10). From this insight, he equated the constant $4\pi^2 k$ *to* GM which turned out to be the law of gravitation.

$$F = -\frac{GMm}{r^2}$$

Again the negative sign in the above equation implies that the gravitational force is attractive.

In the above discussion we assumed that the orbit of the planet to be circular which is not true as the orbit of the planet around the Sun is elliptical. But this circular orbit assumption is justifiable because planet's orbit is very close to being circular and there is only a very small deviation from the circular shape.

UNIT-6(XI-Physics_Vol-2).indd 7 17-08-2020 16:14:57